Normal range of motion (ROM) values

Reference chart · Reviewed by Kevin Lan, OTR/L, ATP · Updated September 2026

Normal range of motion is the number of degrees a healthy joint can move through, from its neutral starting position to its end range. The values below follow the American Academy of Orthopaedic Surgeons (AAOS) normative reference standard used throughout clinical practice — for example, roughly 135° of knee flexion, 150° of elbow flexion, and 180° of shoulder flexion and abduction. Use them as a guide: actual ROM varies with age, sex, and body type. When it's time to write the measurement down, see how to document range of motion.

Printable chart: download the free one-page ROM chart (PDF) — every AAOS value on this page on a single printable sheet for the clinic wall, students, or your clipboard. Free to print and share.

Normal ROM at a glance

Normal range of motion by joint (AAOS) — key movements

JOINT	NORMAL RANGE OF MOTION
Cervical spine	Flexion 45° · extension 45° · lateral flexion 45° · rotation 60°
Shoulder	Flexion 180° · extension 60° · abduction 180° · external rotation 90° · internal rotation 70°
Elbow & forearm	Flexion 150° · extension 0° · pronation 80° · supination 80°
Wrist	Flexion 80° · extension 70° · radial deviation 20° · ulnar deviation 30°
Hand & fingers	MCP flexion 90° · PIP flexion 100° · DIP flexion 90° · thumb MCP 50° · thumb IP 80°

JOINT	NORMAL RANGE OF MOTION
Hip	Flexion 120° · extension 30° · abduction 45° · rotation 45° each way
Knee	Flexion 135° · extension 0°
Ankle	Dorsiflexion 20° · plantarflexion 50° · inversion 35° · eversion 15°

Full per-movement tables for each joint follow below, with links to step-by-step measurement guides.

Measure these angles on your iPhone. Guided placement for 55 movements, with these AAOS normals beside every reading — free to measure.

Get the free app

Cervical spine Shoulder Elbow & forearm Wrist Hand & fingers Hip Knee Ankle

Cervical spine (neck) ROM

Normal cervical spine range of motion (AAOS)

MOVEMENT	NORMAL ROM
Flexion	45°
Extension	45°
Lateral flexion (each side)	45°
Rotation (each side)	60°

How to measure cervical (neck) range of motion — the inclinometer method, technique, and evidence.

Shoulder ROM

Normal shoulder range of motion (AAOS)

MOVEMENT	NORMAL ROM
Flexion	180°
Extension	60°
Abduction	180°
Adduction	40°
External (lateral) rotation	90°
Internal (medial) rotation	70°

How to measure shoulder range of motion — flexion, extension, abduction, and rotation technique.

Elbow & forearm ROM

Normal elbow and forearm range of motion (AAOS)

MOVEMENT	NORMAL ROM
Elbow flexion	150°
Elbow extension	0°
Forearm pronation	80°
Forearm supination	80°

How to measure elbow range of motion — flexion, extension, and forearm rotation technique.

Wrist ROM

Normal wrist range of motion (AAOS)

MOVEMENT	NORMAL ROM
Flexion	80°
Extension	70°
Radial deviation	20°
Ulnar deviation	30°

How to measure wrist range of motion — the most reliable alignment technique, flexion, extension, and deviation.

Hand & finger ROM

Finger and thumb norms are for flexion; full extension returns each joint to a 0° neutral. Total Active Motion (TAM) sums a digit's MCP, PIP, and DIP flexion minus any extension lag. Small digit joints are usually read with a dedicated finger goniometer rather than the full-size universal instrument.

Normal finger and thumb range of motion (AAOS)

JOINT	NORMAL FLEXION
Finger MCP (index–little)	90°
Finger PIP (index–little)	100°
Finger DIP (index–little)	90°
Thumb MCP	50°
Thumb IP	80°

How to measure finger and hand range of motion — finger goniometry, Total Active Motion (TAM), and technique.

Hip ROM

Normal hip range of motion (AAOS)

MOVEMENT	NORMAL ROM
Flexion	120°
Extension	30°
Abduction	45°
Adduction	30°
External rotation	45°
Internal rotation	45°

How to measure hip range of motion — flexion, extension, abduction, adduction, and rotation technique with landmarks.

Knee ROM

Normal knee range of motion (AAOS)

MOVEMENT	NORMAL ROM
Flexion	135°
Extension	0°

How to measure knee range of motion — goniometer landmarks, step-by-step technique, and the reliability evidence.

Ankle ROM

Normal ankle range of motion (AAOS)

MOVEMENT	NORMAL ROM
Dorsiflexion	20°
Plantarflexion	50°
Inversion	35°
Eversion	15°

How to measure ankle range of motion — dorsiflexion and plantarflexion technique with landmarks.

How these values are measured

Each value is the arc from the joint's anatomical neutral (0°) to its end range, measured with a goniometer, an inclinometer, or a smartphone goniometer app. Reliability depends more on technique than on the instrument: standardize the patient's position, align to the same landmark every time, zero at true neutral, and read only a steady end-range value. Smartphone inclinometry has been shown valid and reliable against the universal goniometer in peer-reviewed research. If you are choosing an instrument, digital goniometers and smartphone apps differ mainly in cost and workflow rather than in the underlying measurement, and the same criteria apply when choosing a goniometer app.

Measure these on your iPhone. Goniometer guides you through 55 movements with animated placement and shows the AAOS normal range beside every reading — free to measure. Readings save to patient profiles and export as PDF or CSV reports, and patient data stays encrypted on the device. Download Goniometer on the App Store.

On a computer? Scan with your iPhone camera to get Goniometer.

Frequently asked questions

What is a normal range of motion?

Normal range of motion (ROM) is how far a healthy joint moves, measured in degrees from neutral to end range. The AAOS reference standard puts knee flexion near 135°, elbow flexion near 150°, and shoulder flexion and abduction near 180°. Norms are a guide — actual values vary with age, sex, and body type.

What is normal knee range of motion?

What is normal shoulder range of motion?

What is normal cervical (neck) range of motion?

What is normal elbow range of motion?

How is joint range of motion measured?

Is there a printable normal ROM chart?

References

- American Academy of Orthopaedic Surgeons. *Joint Motion: Method of Measuring and Recording*. Chicago: AAOS; 1965. Normative range-of-motion reference values, as used throughout the Goniometer app.
- Norkin CC, White DJ. *Measurement of Joint Motion: A Guide to Goniometry*. 5th ed. Philadelphia: F.A. Davis; 2016.
- Keogh JWL, Cox A, Anderson S, et al. Reliability and validity of clinically accessible smartphone applications to measure joint range of motion: A systematic review. *PLOS ONE*. 2019;14(5):e0215806. doi:10.1371/journal.pone.0215806

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